NSCLC: Multi-classification of subtypes using Gene expression from RNA-sequence

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Abstract

Lung cancer is the number one cause of mortality worldwide. Early detection of lung cancer is a challenging mission. Non-small cell lung cancer (NSCLC) is the most prevalent subtype of lung cancer, with adenocarcinoma (ACC) and squamous cell carcinoma (SCC) being the most common and large cell lung carcinoma (LCLC) being rarer. Differentiating between several NSCLC subtypes is important for making the right decision about a treatment plan for the patient. The main goal of this study is the development of predictive models combining supervised machine learning and feature selection algorithms to identify at least one gene having the capability of discerning among the different subtypes of lung cancer: ACC, SCC, and LCLC. Principal Component Analysis (PCA) provided a reduced and specific gene signature that allows identifying the subtype of lung cancer of new samples and using SHAP to extract the potential indicator genes. The predictive model of the decision tree performed with 63.39% balanced accuracy; F1 scores of ACC, SCC, and LCLC are 0.79, 0.81, and 0.40, respectively. Finally, CFTR and TNMD may be able to target subtypes of lung cancer in SCC and LCLC, correspondingly.

Introduction

Non-small cell lung cancer (NSCLC) accounts for approximately 85% of all lung cancer cases, making it the most prevalent subtype of lung cancer globally [3]. Lung cancer remains the leading cause of cancer-related deaths, with over 2.2 million new cases diagnosed worldwide each year. NSCLC patients often experience a significant reduction in quality of life due to respiratory symptoms and fatigue, and treatment costs in the U.S. alone exceed $13 billion annually [3,4].

Environmental and lifestyle factors play a major role in NSCLC development. Tobacco smoking remains the leading cause, responsible for nearly 90% of lung cancer deaths. Furthermore, it has been demonstrated that airborne particulate matter (PM2.5) increases the motility and proliferation of cancer cells in NSCLC [5]. Chronic exposure to PM2.5 has been associated with a higher incidence of lung cancer and a poorer prognosis, particularly in urban or industrial areas. This highlights the significance of environmental health in preventing cancer.

Studies using gene expression data and machine learning models such as logistic regression, support vectors, random forest, now achieve prediction accuracies of over 99%

in classifying NSCLC subtypes [6,8,9]. Moreover, the combination of CatBoostClassifier and SHAP was the best-performing machine learning model for predicting on transcriptomic data [9].

This study will use a machine learning model to classify the subtype of NSCLC based on gene expression of the tumor and provide a potential indicator for each subtype to help with personalized diagnostics and treatment planning in NSCLC.

Experimental Methods

Data is collected from the public repository (GSE81089) at the NCBI/Gene Expression Omnibus (GEO) [1,2]. This dataset has 199 samples from the different subtypes of lung cancer addressed in the study, including SCC, AC unspecified, and LCLC/NOS. The gene information from the microarray is illustrated in normalized fragments per kilobase million (FPKM) format.

In the data pre-processing step using Python, the gene ID was converted into a gene name based on Homo sapiens. GRCh38.77 in order to interpret and then statistically analyze. The dataset is imbalanced, with 67 samples of SCC, 108 samples of AC unspecified, and 24 samples of LCLC/NOS, and then separated into a training set (159 samples) and a testing set (40 samples).

Feature reduction

Due to high-dimensional genomic data, 3 alternative approaches were compared by integrating a temporary classifier with each of the feature reduction algorithms proposed. There are t-distributed Stochastic Neighbor Embedding (t-SNE) [17], Uniform Manifold Approximation and Projection (UMAP) [18], and Principal Component Analysis (PCA) [19], measuring for each integration the balanced accuracy on both the training and test datasets.

Handling imbalanced data

To find the best strategy for handling an imbalanced dataset, there are 3 simple techniques: computing class weight, Synthetic Minority Over-sampling Technique (SMOTE), and undersampling. The class weight is calculated within the sklearn library (utils.class_weight), computing based on inverse frequency among classes. SMOTE, the imbalanced learn function, is considered a cornerstone for contemporary imbalanced data resampling [15], having inspired numerous other approaches based on the idea of interpolating nearest minority class observations during the oversampling. These methods typically focus the oversampling in a specific region, modifying the original class distribution in the process [14]. For the undersampling technique, the RandomUnderSampler function of the imbalanced learn, which randomly picks the majority class, is used in this study [16]. Compare these techniques by integrating them into the temporary prediction model and assessing them with balanced accuracy.

Training and Validation of the Predictive Models

After setting up the appropriate feature reduction algorithm and handling the imbalanced data technique, the 5-fold global process was performed on the available training set with the CatBoostClassifier model, which uses gradient boosting on decision trees, and evaluated on the test set regarding balance accuracy and F1 score.

Moreover, through the use of SHAP, which computes Shapley values to provide information about important features and model interpretation [9].

Results and Discussion

This section shows the results of the simulations proposed for this study. As previously stated in the Methods section, different integrations of feature reductions and handling imbalanced data techniques were considered and compared. First, results using PCA for feature reduction provided the best performance at 44.43% balanced accuracy, as shown in Figure 1. Furthermore, the experiments on the number of the components in Figure 2 are thus supported by the performance of the temporary classifiers, which resulted in using 25 components containing the maximum information to discern among the address classes with 69.39% balanced accuracy.

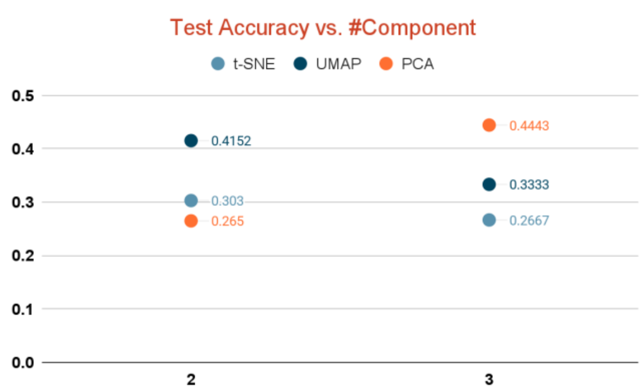

Figure 1: Test accuracy comparison of feature reductions

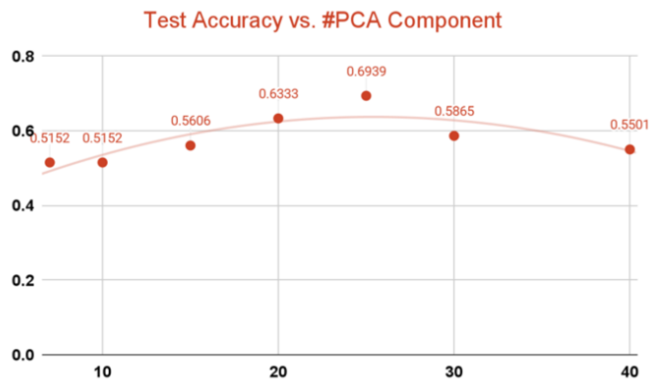

Figure 2: Test accuracy comparison among the numbers of PCA component

Secondly, the result of comparing imbalanced handling techniques shows in Figure 3. With class weight method gave the best performance at 63.33% balanced accuracy.

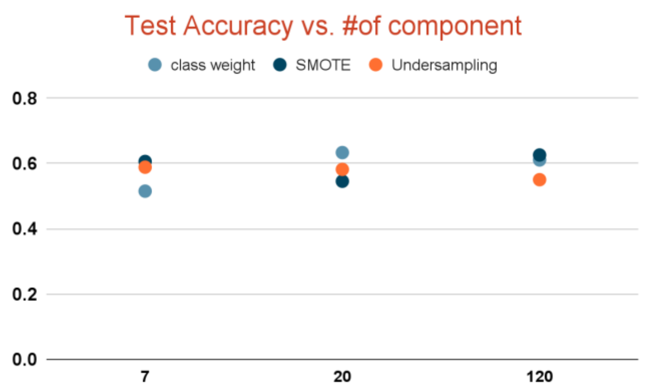

Figure 3: Test accuracy comparison among data imbalanced techniques

Classification Performance

The CatBoostClassifier model was trained using 5-fold cross-validation and class weighting to handle class imbalance. Feature reduction via PCA was performed, with 25 components yielding the best performance. Figure 4 shows the symmetric decision tree, tree depth = 3, obtained by the CatBoostClassifier for the three different classes of classification on the training dataset. On the test set, the model achieved a balanced accuracy of 69.39% shown in Figure 5, with the following F1-scores across NSCLC subtypes:

- Squamous cell carcinoma (SCC): 0.81
- AC unspecified: 0.79
- Large cell carcinoma (LCLC/NOS): 0.40

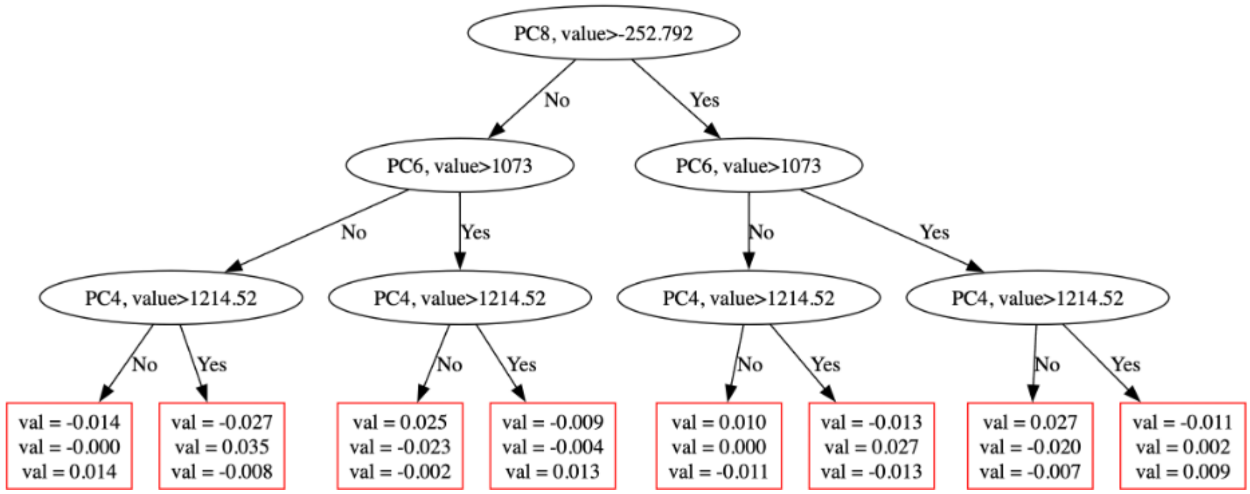

Figure 4: the symmetric decision tree obtained by the CatBoostClassifier

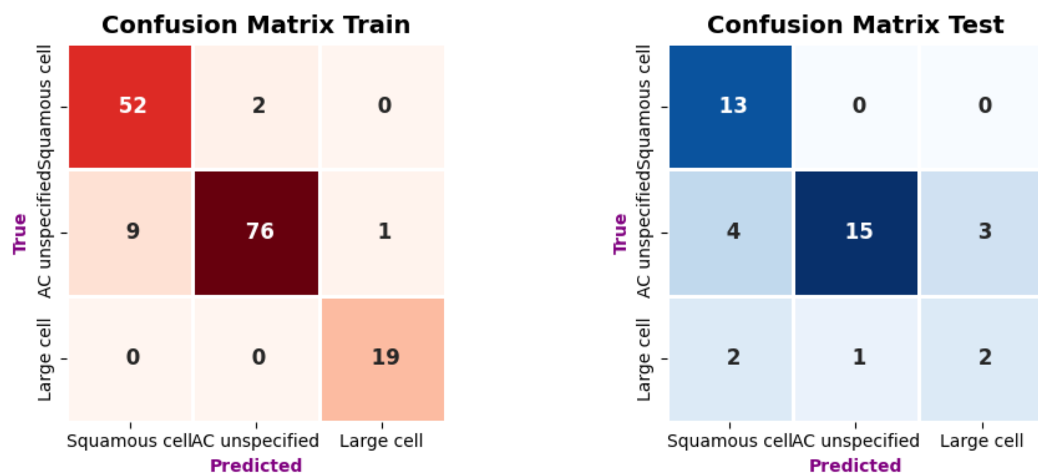

Figure 5: Confusion matrixes of training and testing data

The low F1-score for the LCLC subtype highlights the challenge of predicting underrepresented classes in imbalanced datasets, while SCC subtype and AC unspecified have potential ability to classify.

Model Interpretation

Based on the absolute mean of Shapley values of 25 PCA components, the interpretation of individual genes belongs to the model predictions among the top 3 significant components shown in Figure 6. Across all subtypes, genes such as SCYL3, C1orf112, FGR, TSPAN6, and FUCA2 were consistently ranked among the top features.

Two genes of special clinical relevance emerged shown as Figure 7:

- CFTR (Cystic Fibrosis Transmembrane Conductance Regulator) was strongly associated with SCC subtype. Previous studies link CFTR downregulation with metastasis and poor prognosis in NSCLC patients [10].
- TNMD (Tenomodulin) was prominent in the LCLC subtype. It is believed to mediate tumor-immune interactions and has been linked to cancer metastasis including lung cancer [11].

Figure 6: Top 10 genes expression among classes based on the top 3 PCA components

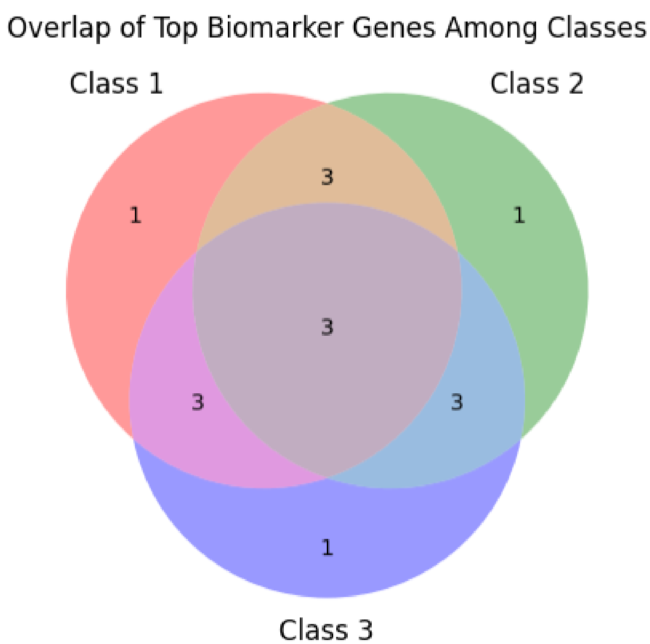

Figure 7: Venn diagram of the top 10 genes among classes; class 1=squamous cell cancer, class 2=AC unspecified, class 3=Large cell/ NOS

Comparison to Prior Work

While prior studies on similar datasets achieved very high accuracy (up to 99%) using logistic regression or CatBoost on dimension-reduced features (e.g., DEGs, RFE) [7,8], this study focused on multi-class classification rather than binary, and on interpretability via SHAP. The lower accuracy reflects a more complex classification problem and a more realistic data distribution.

Clinical Implications

Despite falling short of the desired 80% accuracy threshold, this work demonstrates the potential for subtype-specific biomarker discovery through explainable AI. Gene markers like CFTR and TNMD offer a promising direction for diagnosis refinement and personalized treatment planning in NSCLC. However, additional data—particularly for large cell carcinoma—is necessary to improve performance and confidence.

For further research, expanding the dataset, especially the underrepresented LCLC subtype, to address the imbalanced dataset or adding feature selection techniques such as DEGs or ensemble methods may help improve the model performance. As well as investigate deep learning models like GeneXNet [6] for improved accuracy while preserving interpretability.

Conclusions

This study explored the application of machine learning to classify subtypes of NSCLC based on gene expression profiles from RNA-seq data. Through thoughtful data preprocessing, dimensionality reduction using PCA, and model optimization with CatBoostClassifier integrating class weight, address the challenges posed by the high dimensionality and class imbalance of the dataset.

While the model achieved moderate performance, particularly struggling with the small sample size, SHAP analysis provided interpretability to the model of gene indicators for each subtype, such as CFTR and TNMD, as potential biomarkers for these NSCLC subtypes.

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